

HO�F–RIESZ: Round 001

Smooth finite-particle identities and initial obstructions

Campaign research memorandum

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Baseline: 475a5399828bc6e2ccbade08c59b8778638df14a. This memorandum records the first research round. The singular subcritical and critical fluctuation theorems remain open. Exact smooth identities, bounded analytic estimates, counterexamples, and source claims carry separate statuses. The final round report gives the audit dispositions and exact input hashes. The one-body and raw pair calculations independently reconstruct the smooth versions of Facts 1.4 and 1.2 in the supplied note, pages 1–2; the expanded pair operator and all contractions are derived explicitly in this round.

1 Frozen normalization

On the unit torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$, let g be smooth, real, even and of mean zero, let $K = -\nabla g$, $b = -\nabla V$, and $\nu = \beta_N^{-1}$. For $N \geq 2$,

$$dx_i = \left(b(x_i) + \frac{1}{N} \sum_{j \neq i} K(x_i - x_j) \right) dt + \sqrt{2\nu} dW_i.$$

The smooth, positive reference density solves $\partial_t \mu = -\operatorname{div}(u\mu) + \nu \Delta \mu$, where $u = b + K * \mu$, on a fixed $[0, T]$. Its dependence on N , temperature and regularization is retained. Put $\eta = N^{-1} \sum_i \delta_{x_i}$ and $\rho = \eta - \mu$. All particle tuples below are ordered with distinct labels; coincidences of positions are harmless for smooth kernels. Denominators are N^k , not falling factorials. For a symmetric pair kernel,

$$\mathcal{U}_2[\Phi] = \frac{1}{N^2} \sum_{i \neq j} \Phi(x_i, x_j) - \frac{2}{N} \sum_i \int \Phi(x_i, y) \mu(y) dy + \iint \Phi(x, y) \mu(x) \mu(y) dx dy.$$

We write $P[\Phi] = \frac{1}{2} \mathcal{U}_2[\Phi]$. For $k \geq 0$, $\mathcal{U}_k$ is the alternating sum over subsets of slots: insert distinct particle labels in that subset, integrate the other slots against μ , and multiply by $(-1)^{k-|A|} N^{-|A|}$. This is background centering, not exact expectation centering: under iid μ data, $\mathbb{E} \mathcal{U}_2[\Phi] = -N^{-1} \mu^{\otimes 2}(\Phi)$.

The three Riesz regimes remain separate:

$$\beta_N N^{2s/d-1} \rightarrow 0, \quad \lambda_N := \beta_N N^{s/d-1} \rightarrow 0, \quad \lambda_N \rightarrow \lambda \in (0, \infty).$$

The fluctuation scale is $\sigma_N = \min(\sqrt{N\beta_N}, \kappa_N)$; for iid preparation $\kappa_N = \sqrt{N}$. No fluctuation theorem is inferred from these definitions. For a fixed smooth kernel the Riesz exponent alone does not determine microscopic power counting.

2 One-body and pair identities

Set

$$\begin{aligned} Af &= u \cdot \nabla f + \nu \Delta f, & Rf(y) &= \int K(x-y) \cdot \nabla f(x) \mu(x) dx, \\ Lf &= Af + Rf, & J_f(x, y) &= (\nabla f(x) - \nabla f(y)) \cdot K(x-y). \end{aligned}$$

Then direct expansion of the particle generator gives

$$d\rho(f) = \left\{ \rho((\partial_t + L)f) + \frac{1}{2} \mathcal{U}_2[J_f] \right\} dt + dM_f, \quad dM_f = \frac{\sqrt{2\nu}}{N} \sum_i \nabla f(x_i) \cdot dW_i. \quad (1)$$

In particular the backward equation $(\partial_t + L)f = 0$ yields exact duality with the time-integrated quadratic term retained. To prove (1), expand $\eta = \mu + \rho$ in $\int \eta(dx) (K * \rho)(x) \cdot \nabla f(x)$. The background part is $\rho(Rf)$, and oddness of K symmetrizes the remaining term. Since $K(0) = 0$, the particle diagonal contributes zero to J_f .

For symmetric Φ , define

$$\begin{aligned} R_x \Phi(x, y) &= \int K(z-x) \cdot \nabla_z \Phi(z, y) \mu(z) dz, \\ R_y \Phi(x, y) &= \int K(z-y) \cdot \nabla_z \Phi(x, z) \mu(z) dz, \\ L_2 &= A_x + A_y + R_x + R_y, \\ B\Phi(x, y) &= K(x-y) \cdot (\nabla_x - \nabla_y) \Phi(x, y), \\ C\Phi(x, y, z) &= \text{Sym}_3 [K(x-z) \cdot \nabla_x \Phi(x, y)]. \end{aligned}$$

Here symmetrization is the average over all six permutations. The exact pair identity, in the $P = \mathcal{U}_2/2$ convention, is

$$\begin{aligned} dP[\Phi] &= \left\{ P[(\partial_t + L_2 + N^{-1}B)\Phi] + \mathcal{U}_3[C\Phi] \right. \\ &\quad \left. + N^{-1} \rho((B\Phi)_\mu) + (2N)^{-1} \mu^{\otimes 2}(B\Phi) \right\} dt + dM_\Phi. \end{aligned} \quad (2)$$

The response operators R_x, R_y cannot be omitted from a corrector operator. The internal term B/N and both lower contractions are explicit finite-particle effects.

2.1 Direct identity and diagonal accounting

Let

$$H_i[\Phi] = \frac{1}{N} \sum_{j \neq i} \nabla_x \Phi(x_i, x_j) - \int \nabla_x \Phi(x_i, y) \mu(y) dy, \quad F_i = (K * \rho)(x_i).$$

Applying Itô's formula to the ordered pair sum and to its two background terms gives the raw identity

$$dP[\Phi] = \left\{ P[(\partial_t + A_x + A_y)\Phi] + \frac{1}{N} \sum_i F_i \cdot H_i[\Phi] \right\} dt + dM_\Phi.$$

There are no cross Brownian terms between distinct particle labels. Equivalently, start with the full product $\frac{1}{2} \rho^{\otimes 2}(\Phi)$ and subtract $\eta(\Phi_\Delta)/(2N)$, where $\Phi_\Delta(x) = \Phi(x, x)$. The full-product trace is $\nu N^{-1} \eta(\text{tr } D_{xy} \Phi(x, x))$.

The chain rule

$$\Delta\Phi_\Delta(x) = (\Delta_x + \Delta_y + 2 \operatorname{tr} D_{xy})\Phi(x, x)$$

cancels this trace exactly in the deleted statistic. This cancellation does not remove the martingale. Expansion of the remaining nonlinear force term into distinct triples and the coincident internal label gives (2); complete independent derivations are preserved in the accompanying algebra and BBGKY reports.

2.2 Martingales

The pair martingale and brackets are

$$\begin{aligned} dM_\Phi &= \frac{\sqrt{2\nu}}{N} \sum_i H_i[\Phi] \cdot dW_i, \\ \frac{d[M_\Phi, M_\Psi]}{dt} &= \frac{2\nu}{N^2} \sum_i H_i[\Phi] \cdot H_i[\Psi], \\ \frac{d[M_f, M_\Phi]}{dt} &= \frac{2\nu}{N^2} \sum_i \nabla f(x_i) \cdot H_i[\Phi]. \end{aligned}$$

Every limiting corrector argument must control these terms and their cross variations, not merely the drift.

3 Verification scope

An independent exact rational Laurent-polynomial calculation compares particle-coordinate differentiation with both the raw and decomposed pair identities. Ninety tests cover $N = 2, 3, 4$, diffusivities $0, 1/3, 2$, and five symmetric trigonometric kernels, at homogeneous reference and a one-mode smooth interaction. All coefficient residuals vanish exactly. The zero-diffusion row is a limiting diagnostic. This battery is supporting evidence, not a proof of the general identities. The code, command, hashes and limitations are recorded in certificate CMP-001.

4 Exact smooth all-order recursion

For a symmetric k -slot kernel define

$$\begin{aligned} C_k \Phi &= \sum_{1 \leq a < b \leq k} K(x_a - x_b) \cdot (\nabla_a - \nabla_b) \Phi, \\ A_k^+ \Phi &= \operatorname{Sym}_{k+1} \sum_{a=1}^k K(x_a - x_{k+1}) \cdot \nabla_a \Phi(x_1, \dots, x_k), \\ Q_k \Phi(x_1, \dots, x_{k-1}) &= \int \sum_{a=1}^{k-1} K(x_a - y) \cdot (\nabla_a - \nabla_y) \Phi(x_1, \dots, x_{k-1}, y) \mu(dy), \\ R_k \Phi(x_1, \dots, x_{k-2}) &= \iint K(y - z) \cdot (\nabla_y - \nabla_z) \Phi(x_1, \dots, x_{k-2}, y, z) \mu(dy) \mu(dz). \end{aligned}$$

Let L_k be the sum of the local and response operators in each slot, using the integrated-variable convention already fixed for L_2 . The finite-subset proof gives

$$d\mathcal{U}_k[\Phi] = \left\{ \mathcal{U}_k[(\partial_t + L_k + N^{-1}C_k)\Phi] + \mathcal{U}_{k+1}[A_k^+\Phi] + \frac{k}{N}\mathcal{U}_{k-1}[Q_k\Phi] + \frac{\binom{k}{2}}{N}\mathcal{U}_{k-2}[R_k\Phi] \right\} dt + dM_k[\Phi]. \quad (3)$$

For $k = 1$ both lower terms are absent. At $k = 2$, $A_2^+ = 2C$ and (3) reduces to twice (2). The martingale is exactly the particle-coordinate gradient of the finite statistic:

$$dM_k[\Phi] = \sqrt{2\nu} \sum_{i=1}^N \nabla_{x_i} \mathcal{U}_k[\Phi] \cdot dW_i, \quad \frac{d[M_k[\Phi], M_\ell[\Psi]]}{dt} = 2\nu \sum_i \nabla_{x_i} \mathcal{U}_k[\Phi] \cdot \nabla_{x_i} \mathcal{U}_\ell[\Psi].$$

Each gradient differentiates the explicit ordered-subset definition, so all label coincidences between the two factors remain included. The general proof and expanded bracket contractions are preserved in the completed recursion report. An independently written particle-coordinate verifier passes 144 exact rational Fourier cases through $k = 4$, including $k > N$. This finite battery alone does not certify the all-order assertion.

Neither the finite coefficients in (3) nor a fixed smooth kernel determines the singular critical size of the terms. That question also involves partial diagonals, the law, the corrector norms, and the number of levels. The critical power-counting and summability gate stays open until those estimates are proved.

5 A bounded smooth pair estimate

Fix an integer $m \geq 2$ and use the maximum of the componentwise derivatives through order m as the C^m norm. Write

$$\kappa_m = \|K\|_{C^m}, \quad b_m = \|b\|_{C^m}, \quad M_1(t) = \sum_{a=1}^d \|\partial_a \mu_t\|_{L^1},$$

$$A_m = b_m + \frac{3}{2}\kappa_m, \quad c_m(t) = 2d(2^m - 1)A_m + 2(d\kappa_m + \kappa_0 M_1(t)).$$

For every $N \geq 2$, $\nu \geq 0$ and continuous-time smooth symmetric source F , the equation

$$(\partial_t + L_2 + B/N)\Phi = -F, \quad \Phi_T = 0$$

has a unique smooth symmetric solution in the mild C^m class, and

$$\|\Phi_t\|_{C^m} \leq \int_t^T \exp\left(\int_t^r c_m(a) da\right) \|F_r\|_{C^m} dr. \quad (4)$$

Every background dependence is displayed. No bound on parameter-dependent M_1 or the one-body backward test is inferred from its notation. For $F = J_f$, $\|J_f\|_{C^m} \leq d2^{m+1}\kappa_m\|f\|_{C^{m+1}}$.

Here is the decisive derivative estimate. For $\partial_x^\alpha \partial_y^\gamma R_x \Phi$, if $\alpha = 0$, integrate the z derivative against $\text{div}(K(z-x)\mu(z))$. If $|\alpha| \geq 1$, leave that derivative on Φ and put the x derivatives on K ; then $|\gamma| + 1 \leq m$. Thus

$$\|(R_x + R_y)\Phi\|_{C^m} \leq 2(d\kappa_m + \kappa_0 M_1)\|\Phi\|_{C^m}.$$

Include B/N in the drift $a^N = (u(x) + K(x - y)/N, u(y) - K(x - y)/N)$, whose C^m norm is at most A_m . The additive-noise flow gives a propagator with bounds independent of diffusivity; a time-ordered Volterra series for the bounded response constructs the solution. Independently, differentiation of the PDE and the maximum principle bound its drift commutator by $2d(2^m - 1)A_m\|\Phi\|_{C^m}$. Gronwall gives (4). These two complete mechanisms and their explicit constants appear in the smooth memorandum. Three solvable tests and 120 exact rational Fourier checks include zero and large diffusivity, negative interaction amplitude, and nonzero internal transport. The analytic audit is separate from these same-context proofs.

This does not yield a Riesz cutoff-uniform bound. For $\hat{g}_\varepsilon(k) = c_{d,s}|k|^{s-d}e^{-4\pi^2\varepsilon|k|^2}$, $0 < s < d$, positive $c_{d,s}$ and $k \neq 0$, lattice summation gives

$$\|K_\varepsilon\|_{C^m} \leq C\varepsilon^{-(s+m+1)/2}, \quad DK_\varepsilon(0) = a_\varepsilon I, \quad a_\varepsilon \asymp \varepsilon^{-(s+2)/2}.$$

For the fixed test $f_T(x) = \cos(2\pi x_1)$ and local coordinates $x = q + r/2$, $y = q - r/2$,

$$\partial_{r_1}^2 J_{f_T, \varepsilon}(q + r/2, q - r/2) \Big|_{q=r=0} = -8\pi^2 a_\varepsilon.$$

The C^2 source norm therefore diverges. In particular a single bound $\|\Phi_{T-\tau}\|_{C^2} \leq C\tau$ for every cutoff and all small τ is false. This leaves weaker norms and justified joint limits open.

6 Two explicit route obstructions

On $\mathbb{T}^1$, $0 < s < 1$, the prescribed periodic Riesz kernel has $g_s(r) = |r|^{-s} + H_s(r)$ locally, with H_s smooth. Choose smooth unit-mass nonnegative bumps p, q on disjoint nearby arcs where $g_s(x - y) > 0$, and set

$$w(x, y) = p(x)q(y) + q(x)p(y), \quad f = p - q, \quad h = wg_s.$$

Then $w \geq 0$, $w(x, x) = 0$, and h is smooth. Nevertheless

$$\frac{1}{2} \iint h(x, y) f(x) f(y) dx dy = - \iint p(x)^2 q(y)^2 g_s(x - y) dx dy = -A < 0.$$

For $\mu = 1$, take quantile particles of the density $1 + aN^{-(1-s)/4}f$, with $a > 0$ small and fixed. Their Wasserstein error is at most $1/(2N)$, so their deleted weighted energy is

$$P_N[h] = -Aa^2 N^{-(1-s)/2} + O(N^{-1}).$$

This violates any proposed all-configuration lower bound with zero diagonal main term and error $CN^{s-1-\delta}$, $\delta > 0$, even though the empirical measures converge to μ . The full proof also treats strictly positive weights. Pointwise weight positivity is therefore insufficient; no prepared-law theorem is disproved.

For a general smooth symmetric weight put $d_w(x) = w(x, x)$. The specific remainder

$$D_w(x, y) = (w(x, y) - w(x, x))g'_s(x - y) + \partial_1 w(x, y)g_s(x - y)$$

has expansion

$$D_w(x, x - r) = \frac{1+s}{2} d'_w(x) |r|^{-s} + O(|r|^{1-s}) + O(1).$$

For $w = 2 + \frac{1}{2}(\sin(2\pi x) + \sin(2\pi y))$, at $x = 0$ its ratio to $g_s(r)$ tends to $\pi(1+s)$. It generically retains the original singularity. A constant diagonal eliminates this leading term and gives $D_w = O(|r|^{1-s})$. Complete witnesses, quantifiers and independent normalization calculations are in the falsification memorandum.

7 Source boundaries and the remaining assertion

The seven-page imported note remains immutable. Its labels *Fact*, *Hope*, *Missing ingredient* and *Obstruction* are source labels, not campaign certifications. Its finite-temperature renormalized-energy formula cannot omit the energy martingale. The pointwise comparison $g/2 \leq g_\theta \leq 3g/2$ cannot hold where the zero-mean periodic kernel is negative. Neither issue disproves the scientific mission. The source ledger distinguishes version-locked primary normalization from the unavailable private source and its unused claims.

For the one-dimensional circular Hamiltonian $H_N = N^{-s} \sum_{i \neq j} g_s(x_i - x_j)$ in Boursier's *Optimal local laws and CLT for the circular Riesz gas*, arXiv:2112.05881v5, equations (1.1), (1.4), the campaign Gibbs energy is $(2N)^{-1} \sum_{i \neq j} g_s$. Consequently the inverse temperature in that source equals $\beta_N N^{s-1}/2 = \lambda_N/2$. Fixed source temperature belongs to the campaign's critical scale. No temperature-uniform or nonequilibrium import follows. The ordered-source report records the exact versions, theorem locations, a printed Fourier-normalization discrepancy, and the stopped gap identity. Its suspect numerical covariance normalization is quarantined.

For $(\partial_t + L)f = 0$ and the pair solution with source $-J_f$, the exact corrected observable is

$$\begin{aligned} \rho_T(f_T) &= \rho_0(f_0) + P_N[\Phi_0] + M_f(T) + M_\Phi(T) \\ &\quad + \int_0^T \{ \mathcal{U}_3[C\Phi_t] + N^{-1} \rho_t((B\Phi_t)_{\mu_t}) + (2N)^{-1} \mu_t^{\otimes 2}(B\Phi_t) \} dt. \end{aligned}$$

The first open load-bearing probabilistic assertion is a law-specific bound on the integrated cubic term at scale σ_N , together with the initial correction and scaled martingale/cross brackets. The active proof-obligation card first states a fixed-smooth iid problem with visible uniform background/test bounds. The singular extension must additionally control the cutoff and the corrector-order tail. At criticality, surviving terms may require an infinite hierarchy or enlarged state; their vanishing is not assumed. M2's power-counting gate and the singular fluctuation theorems remain open.